Practical No.05

Aggregate Functions

```
mysql> Insert into Student(Roll_no,Name,Marks) Values (01,'Anuj',50),(02,'Simran',85),(03,'Riya',50),(04,'Vedant',80),(05,'Suraj',40);
Query OK, 5 rows affected (0.02 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> Select * from Student;
+---------+--------+-------+
| Roll_no | Name   | Marks |
+---------+--------+-------+
|       1 | Anuj   |    50 |
|       2 | Simran |    85 |
|       3 | Riya   |    50 |
|       4 | Vedant |    80 |
|       5 | Suraj  |    40 |
+---------+--------+-------+
5 rows in set (0.00 sec)
```

```
mysql> Select SUM(Marks) AS Student_Total_marks From Student;
+----------------------+
| Student_Total_marks |
+----------------------+
|                  305 |
+----------------------+
1 row in set (0.00 sec)

mysql> Select Count(*) AS total_records from Student;
+---------------+
| total_records |
+---------------+
|             5 |
+---------------+
1 row in set (0.00 sec)

mysql> Select avg(Marks) As Average_Marks From Student;
+---------------+
| Average_Marks |
+---------------+
|       61.0000 |
+---------------+
1 row in set (0.00 sec)

mysql> Select min(Marks) As Minimum_Marks,max(Marks) As Maximum_Marks From Student;
+---------------+---------------+
| Minimum_Marks | Maximum_Marks |
+---------------+---------------+
|            40 |            85 |
+---------------+---------------+
1 row in set (0.00 sec)
```

String Functions

```
mysql> Select concat('Simran',' ','Bhoir') As Full_name;
+--------------+
| Full_name    |
+--------------+
| Simran Bhoir |
+--------------+
1 row in set (0.00 sec)

mysql> Select char_length('Computer') As String_Length ;
+---------------+
| String_Length |
+---------------+
|             8 |
+---------------+
1 row in set (0.00 sec)

mysql> Select upper('Computer') As Uppercase,lower('Computer') As Lowercase ;
+-----------+-----------+
| Uppercase | Lowercase |
+-----------+-----------+
| COMPUTER  | computer  |
+-----------+-----------+
1 row in set (0.00 sec)

mysql> Select Replace('Welcome to DBMS','DBMS','SQL') AS Updated_String;
+----------------+
| Updated_String |
+----------------+
| Welcome to SQL |
+----------------+
1 row in set (0.00 sec)

mysql> Select reverse('Helloooo') As Reversed_String;
+------------------+
| Reversed_String |
+------------------+
| oooolleH         |
+------------------+
1 row in set (0.00 sec)

mysql> Select concat_ws('$','Hello','from','SQL');
+---------------------------------------+
| concat_ws('$','Hello','from','SQL') |
+---------------------------------------+
| Hello$from$SQL                        |
+---------------------------------------+
1 row in set (0.00 sec)
```

Practical No.06

```
mysql> Select * from Employee;
+--------+----------+-----------+--------+
| Emp_id | Emp_name | Emp_dept  | Salary |
+--------+----------+-----------+--------+
|    101 | Josh     | IT        |  50000 |
|    102 | Ayush    | Sales     |  55000 |
|    103 | Smriti   | Marketing |  60000 |
|    104 | Jemi     | HR        |  50000 |
+--------+----------+-----------+--------+
4 rows in set (0.00 sec)
```

```
mysql> Select * from Department;
+-----------+--------------+
| Emp_dept  | Manager_name |
+-----------+--------------+
| IT        | Rohit        |
| Sales     | Tilak        |
| Marketing | Tina         |
| HR        | Sam          |
| Finance   | Samaira      |
+-----------+--------------+
5 rows in set (0.00 sec)
```

1. Natural Join

```
mysql> Select * From Employee Natural Join Department;
+-----------+--------+----------+--------+--------------+
| Emp_dept  | Emp_id | Emp_name | Salary | Manager_name |
+-----------+--------+----------+--------+--------------+
| IT        |    101 | Josh     |  50000 | Rohit        |
| Sales     |    102 | Ayush    |  55000 | Tilak        |
| Marketing |    103 | Smriti   |  60000 | Tina         |
| HR        |    104 | Jemi     |  50000 | Sam          |
+-----------+--------+----------+--------+--------------+
4 rows in set (0.00 sec)
```

2. Equi Join

```
mysql> Select Employee.Emp_id,Employee.Emp_name,Department.Emp_Dept From Employee,Department Where Employee.Emp_dept = Department.Emp_dept;
+--------+----------+-----------+
| Emp_id | Emp_name | Emp_Dept  |
+--------+----------+-----------+
|    101 | Josh     | IT        |
|    102 | Ayush    | Sales     |
|    103 | Smriti   | Marketing |
|    104 | Jemi     | HR        |
+--------+----------+-----------+
4 rows in set (0.00 sec)
```

3. Self Join

```
mysql> Select e1.Emp_name AS Employee_1,e2.Emp_name AS Employee_2,e1.Emp_dept From Employee As e1 Join Employee as e2 on e1.Emp_dept =e2.Emp_dept and e1.Emp_id < e2.Emp_id;
+-----------+-----------+----------+
| Employee_1 | Employee_2 | Emp_dept |
+-----------+-----------+----------+
| Josh      | Sara      | IT       |
+-----------+-----------+----------+
1 row in set (0.00 sec)
```

4. Inner Join

```
mysql> Select Department.Manager_name,Employee.Emp_name,Employee.Salary From Employee Inner Join Department On Employee.Emp_dept=Department.Emp_dept;
+--------------+----------+--------+
| Manager_name | Emp_name | Salary |
+--------------+----------+--------+
| Rohit        | Josh     |  50000 |
| Tilak        | Ayush    |  55000 |
| Tina         | Smriti   |  60000 |
| Sam          | Jemi     |  50000 |
+--------------+----------+--------+
4 rows in set (0.00 sec)
```

5. Left Join

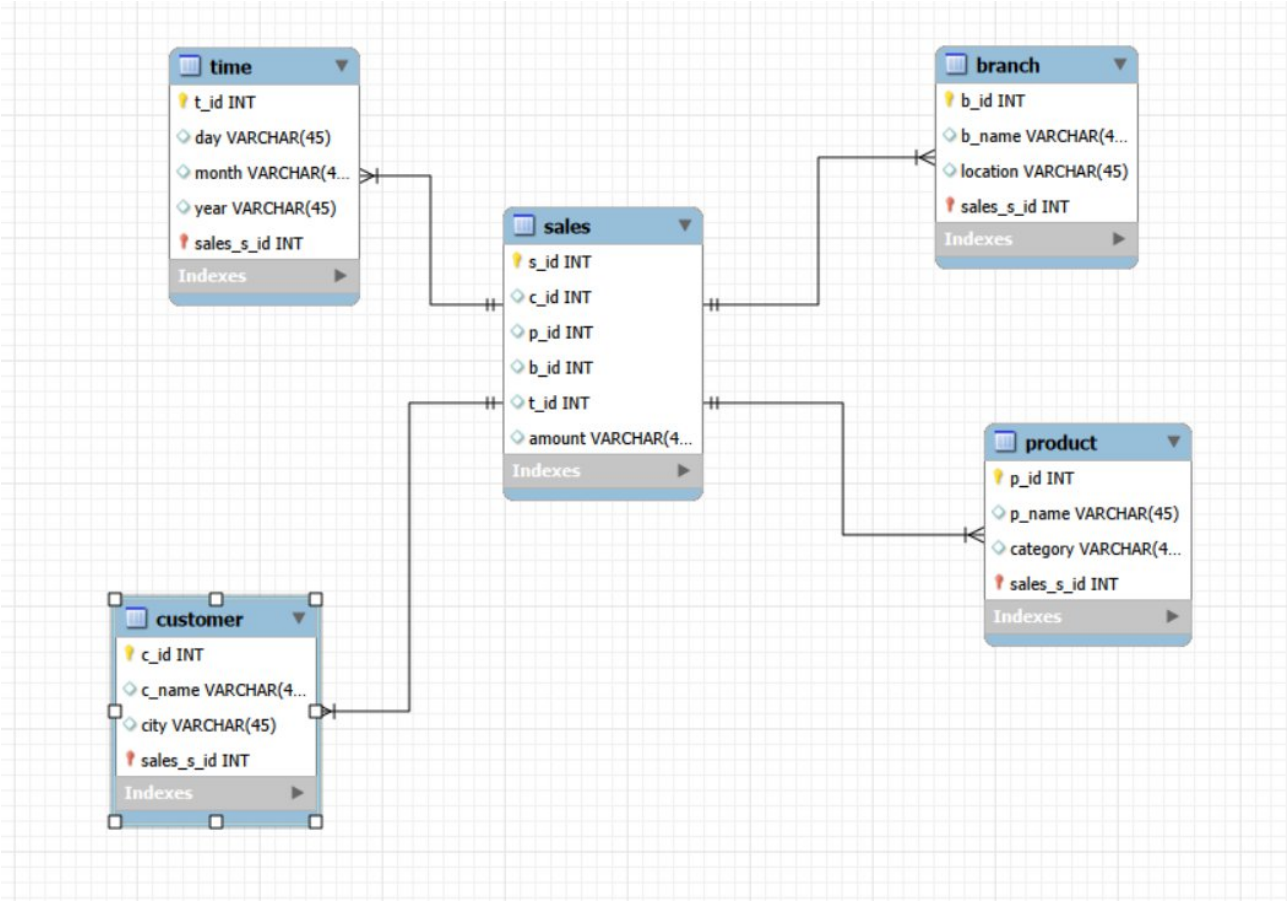
```
mysql> Select Department.Manager_name,Employee.Emp_name,Employee.Salary,Department.Emp_dept From Employee Left Join Department On Employee.Emp_dept=Department.Emp_dept;
+--------------+----------+--------+-----------+
| Manager_name | Emp_name | Salary | Emp_dept  |
+--------------+----------+--------+-----------+
| Rohit        | Josh     |  50000 | IT        |
| Tilak        | Ayush    |  55000 | Sales     |
| Tina         | Smriti   |  60000 | Marketing |
| Sam          | Jemi     |  50000 | HR        |
+--------------+----------+--------+-----------+
4 rows in set (0.00 sec)
```

6. Right Join

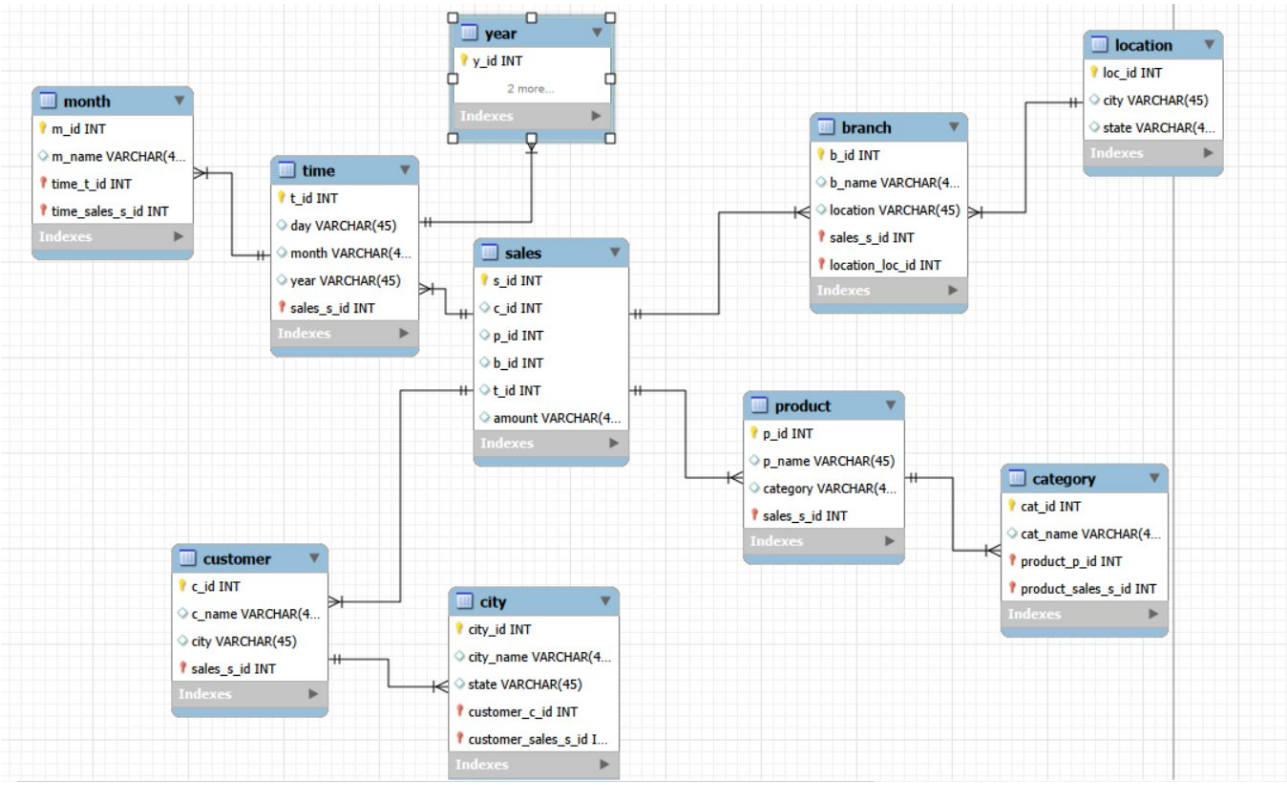
```
mysql> Select Department.Manager_name,Employee.Emp_name,Employee.Salary,Department.Emp_dept From Employee Right Join Department On Employee.Emp_dept=Department.Emp_dept;
+--------------+----------+--------+-----------+
| Manager_name | Emp_name | Salary | Emp_dept  |
+--------------+----------+--------+-----------+
| Rohit        | Josh     |  50000 | IT        |
| Tilak        | Ayush    |  55000 | Sales     |
| Tina         | Smriti   |  60000 | Marketing |
| Sam          | Jemi     |  50000 | HR        |
| Samaira      | NULL     |   NULL | Finance   |
+--------------+----------+--------+-----------+
5 rows in set (0.00 sec)
```

Experiment no 7

Starflake



Snowflake



Practical No.8.B

- **Group by**

```
mysql> SELECT Job, COUNT(*) AS Total_Employees, AVG(Salary) AS Avg_Salary
    -> FROM employee
    -> GROUP BY Job;
+--------------+-----------------+-------------------+
| Job          | Total_Employees | Avg_Salary        |
+--------------+-----------------+-------------------+
| Front-Desk   |               1 |             25000 |
| Manager      |               3 | 43333.333333333336 |
| Housekeeping |               1 |             15000 |
| Accountant   |               1 |             25000 |
| Room Service |               1 |             17000 |
| Kitchen-Staff |               1 |             14000 |
+--------------+-----------------+-------------------+
```

- **Having**

```
mysql> SELECT Job, AVG(Salary) AS Avg_Salary
    -> FROM employee
    -> GROUP BY Job
    -> HAVING AVG(Salary) > 20000;
+------------+-------------------+
| Job        | Avg_Salary        |
+------------+-------------------+
| Front-Desk |             25000 |
| Manager    | 43333.333333333336 |
| Accountant |             25000 |
+------------+-------------------+
3 rows in set (0.00 sec)
```

- **Order by**

```
mysql> SELECT Name, Job, Salary
    -> FROM employee
    -> ORDER BY Salary DESC;
+----------------+---------------+--------+
| Name           | Job           | Salary |
+----------------+---------------+--------+
| Leonardo       | Manager       |  50000 |
| Sanika T.      | Manager       |  50000 |
| John kon       | Manager       |  30000 |
| John Sharma    | Front-Desk    |  25000 |
| Amit Shah      | Accountant    |  25000 |
| Raj Malhotra   | Room Service  |  17000 |
| Sneha Kulkarni | Housekeeping  |  15000 |
| Priya Deshmukh | Kitchen-Staff |  14000 |
+----------------+---------------+--------+
8 rows in set (0.00 sec)
```

Practical No.8.A

- **Single Row Subquery**

```
mysql> SELECT Name, Salary
    -> FROM employee
    -> WHERE Salary > (SELECT AVG(Salary) FROM employee);
+-----------+--------+
| Name      | Salary |
+-----------+--------+
| John kon  |  30000 |
| Leonardo  |  50000 |
| Sanika T. |  50000 |
+-----------+--------+
3 rows in set (0.00 sec)
```

- **Multi-row Subquery**

```
mysql> SELECT Name, Job
    -> FROM employee
    -> WHERE Job IN (SELECT Job FROM employee WHERE Job = 'Manager');
+-----------+---------+
| Name      | Job     |
+-----------+---------+
| John kon  | Manager |
| Leonardo  | Manager |
| Sanika T. | Manager |
+-----------+---------+
3 rows in set (0.00 sec)
```

```
mysql> SELECT Name, Salary
    -> FROM employee
    -> WHERE Salary > ANY (SELECT Salary FROM employee WHERE Job = 'Housekeeping');
+--------------+--------+
| Name         | Salary |
+--------------+--------+
| John Sharma  |  25000 |
| John kon     |  30000 |
| Leonardo     |  50000 |
| Sanika T.    |  50000 |
| Amit Shah    |  25000 |
| Raj Malhotra |  17000 |
+--------------+--------+
6 rows in set (0.00 sec)
```

- **Subquery with Exists**

```
mysql> SELECT e1.Name, e1.Job
    -> FROM employee e1
    -> WHERE EXISTS (
    ->     SELECT 1
    ->     FROM employee e2
    ->     WHERE e2.Job = e1.Job
    ->     AND e2.Salary > 30000
    -> );
+-----------+---------+
| Name      | Job     |
+-----------+---------+
| John kon  | Manager |
| Leonardo  | Manager |
| Sanika T. | Manager |
+-----------+---------+
3 rows in set (0.00 sec)
```

Practical No.10

DCL commands

- **GRANT**

```
mysql> GRANT SELECT,UPDATE ON employee TO 'root'@'localhost';
Query OK, 0 rows affected (0.05 sec)

mysql> flush privileges;
Query OK, 0 rows affected (0.01 sec)

mysql> Select user,host from mysql.user where user='root';
+------+-----------+
| user | host      |
+------+-----------+
| root | localhost |
+------+-----------+
1 row in set (0.01 sec)
```

```
mysql> show grants for 'root'@'localhost';
+-----------------------
```

TCL Commands

- **COMMIT**

```
mysql> UPDATE employee SET Salary =Salary + 5000 Where Job = 'Front-Desk'; COMMIT;
Query OK, 1 row affected (0.08 sec)
Rows matched: 1  Changed: 1  Warnings: 0

Query OK, 0 rows affected (0.00 sec)

mysql> Select * from employee where Job='Front-Desk';
+-------------+------+--------+------------+--------+------------+----------------+----------------+
| Name        | Age  | Gender | Job        | Salary | Phone      | Adhar          | email          |
+-------------+------+--------+------------+--------+------------+----------------+----------------+
| John Sharma |   28 | Male   | Front-Desk |  30000 | 9519563645 | 8569 6523 2541 | john@gmail.com |
+-------------+------+--------+------------+--------+------------+----------------+----------------+
1 row in set (0.07 sec)
```

- **SAVEPOINT**

```
mysql> Update employee Set Salary = Salary +2000 Where Job='Housekeeping'; SAVEPOINT sp1;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

Query OK, 0 rows affected (0.00 sec)

mysql> Select * from employee where Job='Housekeeping';
+----------------+------+--------+--------------+--------+------------+----------------+--------------------+
| Name           | Age  | Gender | Job          | Salary | Phone      | Adhar          | email              |
+----------------+------+--------+--------------+--------+------------+----------------+--------------------+
| Sneha Kulkarni |   32 | Female | Housekeeping |  17000 | 8765432109 | 5678 4523 8526 | sneha.k12@gmail.com |
+----------------+------+--------+--------------+--------+------------+----------------+--------------------+
1 row in set (0.00 sec)
```

- **ROLLBACK**

```
mysql> DELETE FROM employee WHERE Name='John kon'; ROLLBACK;
Query OK, 1 row affected (0.07 sec)

Query OK, 0 rows affected (0.00 sec)

mysql> Select * from employee;
+----------------+------+--------+---------------+--------+------------+----------------+
| Name           | Age  | Gender | Job           | Salary | Phone      | Adhar          |
+----------------+------+--------+---------------+--------+------------+----------------+
| John Sharma    |   28 | Male   | Front-Desk    |  30000 | 9519563645 | 8569 6523 2541 |
| Leonardo       |   25 | Male   | Manager       |  63500 | 8529637415 | 5632 4123 8965 |
| Sanika T.      |   21 | Female | Manager       |  63500 | 4567891235 | 5263 5421 8523 |
| Sneha Kulkarni |   32 | Female | Housekeeping  |  17000 | 8765432109 | 5678 4523 8526 |
| Amit Shah      |   40 | Male   | Accountant    |  25000 | 8523697412 | 8523 1596 2574 |
| Raj Malhotra   |   30 | Male   | Room Service  |  17000 | 9513578426 | 8523 9632 4123 |
| Priya Deshmukh |   29 | Female | Kitchen-Staff |  14000 | 8527413695 | 7412 2589 9632 |
+----------------+------+--------+---------------+--------+------------+----------------+
7 rows in set (0.00 sec)
```

Experiment no 9

OLAP queries:

Slice

[SQL Worksheet]*

```
SELECT *
FROM Revenue
WHERE Branch = 'Branch E';
```

Query result | Script output | DBMS output | Explain Plan | SQL history

Download Execution time: 0.001 seconds

	BRANCH	DEALER	MODEL	CUSTOMER	PROFIT
1	Branch E	Dealer 2	Model Y	Customer 105	19000
2	Branch E	Dealer 2	Model Y	Customer 106	17000
3	Branch E	Dealer 3	Model X	Customer 107	21000

Dice

[SQL Worksheet]*

```
SELECT *
FROM Revenue
WHERE Branch = 'Branch A'
AND Model = 'Model X';
```

Query result | Script output | DBMS output | Explain Plan | SQL history

Download Execution time: 0.008 seconds

	BRANCH	DEALER	MODEL	CUSTOMER	PROFIT
1	Branch A	Dealer 1	Model X	Customer 101	15000

Roll up

[SQL Worksheet]*

```sql
SELECT Branch, Dealer, SUM(Profit) AS Total_Profit
FROM Revenue
GROUP BY ROLLUP (Branch, Dealer);
```

Query result | Script output | DBMS output | Explain Plan | SQL history

Download Execution time: 0.006 seconds

	BRANCH	DEALER	TOTAL_PROFIT
1	Branch A	Dealer 1	15000
2	Branch A	Dealer 2	20000
3	Branch B	Dealer 1	18000
4	Branch E	Dealer 2	36000
5	Branch E	Dealer 3	21000
6	Branch A	*(null)*	35000
7	Branch B	*(null)*	18000
8	Branch E	*(null)*	57000
9	*(null)*	*(null)*	110000

Drill down

[SQL Worksheet]*

```sql
SELECT Branch, Dealer, SUM(Profit) AS Total_Profit
FROM Revenue
GROUP BY Branch, Dealer;
```

Query result | Script output | DBMS output | Explain Plan | SQL history

Download Execution time: 0.006 seconds

	BRANCH	DEALER	TOTAL_PROFIT
1	Branch A	Dealer 1	15000
2	Branch A	Dealer 2	20000
3	Branch B	Dealer 1	18000
4	Branch E	Dealer 2	36000
5	Branch E	Dealer 3	21000

pivot

[SQL Worksheet]*

```sql
SELECT *
FROM (
    SELECT Branch, Model, Profit FROM Revenue
)
PIVOT (
    SUM(Profit)
    FOR Model IN ('Model X' AS ModelX, 'Model Y' AS ModelY)
);
```

Query result | Script output | DBMS output | Explain Plan | SQL history

Download Execution time: 0.006 seconds

	BRANCH	MODELX	MODELY
1	Branch A	15000	20000
2	Branch B	18000	*(null)*
3	Branch E	21000	36000